

# Notebook workflow

Module 2 - Lesson 1

## Overview

Jupyter notebooks combine code, text, and visualizations for exploratory analysis. A clean notebook workflow makes analysis reproducible and shareable.

## Key Concepts

- Notebooks are documents of cells: Markdown explains, code executes, output records results.
- Run cells top-to-bottom; a clean restart-and-run-again proves reproducibility.
- Use Markdown headings, explanations, and conclusions so the notebook reads as a report.
- Keep notebooks for exploration; move production code into .py modules.
- Pin dependency versions and seed random generators for stable results.
- Use version control even for notebooks (review notebooks, not just code).

## Example

```
# 1. Setup
import pandas as pd
import matplotlib.pyplot as plt

# 2. Load data
df = pd.read_csv("courses.csv")

# 3. Explore
print(df.describe())

# 4. Analyze
plt.hist(df["price"])
plt.show()
```

## Summary

Notebooks blend code, prose, and charts for exploration. Clear Markdown, reproducible order, and moving finished logic to modules keep them useful and trustworthy.

## Practice Checklist

- Restart the kernel and run all cells; fix anything that breaks.
- Add Markdown headings and a conclusion to a notebook.
- Extract one repeated analysis into a reusable .py function.